Yiquan Wang

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• Member of IEEE Biometrics Council, Chinese Chemical Society & Biophysical Society of China

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Education

National Base for Mathematical Research and Teaching Talents, Xinjiang University — Urumqi, Xinjiang
B.S. in Mathematics and Applied Mathematics — 2023.09 2027.06

Tsien Excellence in Engineering Program, Tsinghua University & Shenzhen X-Institute — Shenzhen, Guangdong
Joint Program, X Scholar — Representation learning for protein function and dynamics — 2024.06 2027.06

Institute of Neurological and Psychiatric Disorders, Shenzhen Bay Laboratory — Shenzhen, Guangdong
Visiting Student, Wen Yuan's Research Group — Single-cell bioinformatics for neurobiology studies — 2025.07 2025.09

Institute of Systems and Physical Biology, Shenzhen Bay Laboratory — Shenzhen, Guangdong
Visiting Student, Kai Huang's Research Group — IDP, LLPS & genome folding, undruggable-target design — Starting 2026.07

Selected Publications

*Name in bold; † co-first author; * corresponding author.*

Direction 1: Representation Learning & Biomolecular Mechanisms

[1] **Wang, Y.**, et al. (2025). From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction. *Journal of Chemical Information and Modeling*.

[2] **Wang, Y.***, et al. (2026). Sequence-context-aware decoding enables robust reconstruction of protein dynamics from crystallographic B-factors. *Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026*.

[3] **Wang, Y.***, et al. (2026). Integrative transcriptomics, machine learning, and molecular docking derive a DAM-like macrophage signature for risk stratification and therapeutic nomination in glioblastoma. *Computational Biology and Chemistry*.

[4] **Wang, Y.**, et al. (2025). Diffusion Models at the Drug Discovery Frontier: A Review on Generating Small Molecules versus Therapeutic Peptides. *Biology*.

[5] Wang, X.*, **Wang, Y.***, et al. (2025). Octopus Inspired Optimization (OIO): A Hierarchical Framework for Navigating Protein Fitness Landscapes. *In 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE.

[6] **Wang, Y.***, et al. (2026). Comment on: "GLM7 A Novel Composite Glycolipid Index Derived from Routine Health Indicators for Enhanced Diagnosis and Prediction of Multimorbidity". *Advanced Science*. (correspondence).

[7] **Wang, Y.***, et al. (2025). AI for disease prediction: Performance insights and key limitations. *Journal of Clinical Neuroscience*. (correspondence).

Direction 2: First-Principles of Biological Complexity Based on Mathematical Physics

[8] **Wang, Y.*** (2026). Sub-residue sharpness of protein helix-coil transitions reveals a spatial-spectral uncertainty limit. *arXiv preprint arXiv:2602.21787*.

[9] **Wang, Y.*** (2026). Piecewise integrability of the discrete Hasimoto map for analytic prediction and design of helical peptides. *arXiv preprint arXiv:2602.16255*.

[10] **Wang, Y.*** (2026). Structural barriers of the discrete Hasimoto map applied to protein backbone geometry. *arXiv preprint arXiv:2602.13160*.

Direction 3: Interdisciplinary Explorations & Collaborative Synergies

[11] **Wang, Y.**, Zai, J., Liu, Z., Chen, J., & Sabir, E.* (2026). Hamiltonian Connectivity in k-Ary n-Cubes under a Region-Based Fault Model. *Information Sciences*.

[12] **Wang, Y.***, et al. (2026). AI Driven Discovery of Bio Ecological Mediation in Cascading Heatwave Risks. *Physics and Chemistry of the Earth*.

[13] **Wang, Y.***, et al. (2026). Geometric Fragility in Alzheimer's Disease: Probing the Loss of Hippocampal Hierarchical Abstraction via Contrastive Point Cloud Modeling. *In 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE.

[14] Cai, M., . . . , **Wang, Y.***, & Wei, K.* (2025). APOE ε4-Associated Hippocampal Atrophy Trajectories Across the Alzheimer's Disease Continuum: A Systematic Review, Meta-Analysis, and Longitudinal Validation. *Frontiers in Aging Neuroscience*.

Research Projects

Tsinghua University Tsien Excellence in Engineering Program ESRT — 2024.08-2025.10

- **Project:** From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction.
- Motivated by an intuition that protein structure mirrors music—catalytic sites resembling melodic climaxes and β-sheets resembling calmer passages—I encoded amino acid sequences as 2D spectrograms (protein sonification) for function prediction. Ablations across an 18,000-sequence/12-class benchmark showed the 1D→2D representational structure itself, rather than explicit biophysical priors, was the key driver of performance, matching ESM-2 and ProtBERT and reaching 90.44% on the external CARE benchmark.
- Project Repository. Supervised by Prof. Kai Wei.

Chinese Academy of Sciences (CAS) Innovation Practice Training Program — 2024.11 – 2025.09

- **Project:** A Knowledge Graph-based Q&A System for Heatwave Disaster Adaptation.
- Built HeDA, an autonomous synthesis agent that constructs a knowledge graph (34,933 entities, 42,890 relations) from 8,365 publications to map compound-heatwave cascading risks. On a temporally held-out multi-hop benchmark, graph augmentation improved accuracy by

8.1–9.8% over standalone foundation models; topological analysis surfaced a bio-ecological mediation effect, where biological systems act as the primary structural mediators transforming physical hazards into systemic socioeconomic losses.
- Project Repository. Supervised by Researcher Yong Ge.

Provincial Undergraduate Innovation Training Program 2024.03 – 2025.06
- **Project:** Research on the Generation Cyclability of Cartesian Product Graphs and Related Problems.
- Studied Hamiltonicity of the k-ary n-cube Q_n^k, a classical graph-theoretic interconnection topology. Motivated by the fact that real-world faults are spatially clustered rather than independent, I introduced the Region-Based Fault (RBF) model characterizing faults as connected subgraphs, and gave a constructive proof that Q_n^k remains Hamiltonian-connected for odd $k \geq 3,\ n \geq 2$ under RBF conditions, via an adaptive graph-decomposition argument robust well beyond its theoretical guarantees.
- Project Repository. Supervised by Prof. Eminjan Sabir.

National Undergraduate Innovation Program 2025.05 – 2026.12
- **Project:** Conditional Diffusion Model on Copy Number Variation for Alzheimer's Disease Risk Assessment.

Tsinghua University Tsien Excellence in Engineering Program ORIC 2025.12 – 2026.12
- **Project:** Application of Artificial Intelligence in Whole Genome Selection Breeding.

Chinese Academy of Sciences (CAS) Innovation Practice Training Program 2025.11 – 2026.11
- **Project:** Urban Big Data, Multimodal Fusion and Spatial Intelligence Analysis (Second Author).

Professional Experience

Beijing Frontier Research Center for Biological Structure, Tsinghua University Beijing, China
Intern 2025.07 Present
- Developed the Peptide Design Competition Website, the Polypeptide Structure Database and engaged in protein and polypeptide design research. Supervised by Prof. Yafei Yuan and Prof. Tong Wang.

Institute of Software, CAS & Huawei Mindspore Community Remote
Intern 2024.09 2025.03
- Implemented a VGG19-based model for Pollock-style art generation, focusing on fractal and turbulent feature extraction and the creation of NFT-based digital labels. Featured on Huawei's official Wechat.

Awards & Honors

- **China Undergraduate Life Science Contest**: National Second Prize 2026
- **"Challenge Cup" China College Students' Extracurricular Academic Science and Technology Works Competition**: Autonomous Region Second Prize 2026
- **China College Students' E-Commerce "Innovation, Creativity and Entrepreneurship" Challenge**: Autonomous Region Second Prize 2026
- **China College Student Self-improvement Star** 2025
- **UN Global Leadership and ESG Programme $\times$ X-Institute Shenzhen "AI for the Young and the Elderly" Challenge**: Best Innovation Award 2026
- **Smart Chemical City Challenge (Xinjiang Station)**: Second Prize 2025
- **Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)**: National Second Prize 2025
- **SynBio Challenges**: Silver Award $\times$ 2 2025
- **2025 X-Fusion "Global Innovator Fusion Conference"**: Best Poster Award 2025
- **The Mathematical Contest in Modeling (MCM)**: Honorable Mention 2025
- **Alibaba Cloud Tianchi University Student Competition**: National Finals, 17th Place 2024
- **APMCM Asia-Pacific Mathematical Modeling Competition**: National Third Prize 2024

Academic Activities

Peer Reviewer: ICLR/ICML/NeurIPS workshops; Urban Climate; Physics and Chemistry of the Earth; Mini-Reviews in Medicinal Chemistry; Current Science; F1000Research.

Academic Engagement:
- Tsinghua University-Peking University Center for Life Sciences Summer Camp 2025.07
- Shenzhen Bay Laboratory / Shenzhen Medical Academy of Research and Translation Summer Research 2025.07 – 2025.09
- Tsinghua University Tsien Excellence in Engineering Program – Zero One Scholar 2024.06 – 2027.06
- AI Winter School, Brown University Department of Physics 2025.01
- CAAI Artificial Intelligence and Technology Ethics Training Course 2024.09 – 2024.12
- Fudan University Summer School of Mathematical Logic 2024.08
- Wuhan University National Tianyuan Mathematics Center Discussion Class 2024.03 – 2024.06

Website Development

- **Peptide Design Competition, Tsinghua University** — [Link]
- **Polypeptide Structure Database, Tsinghua University** — [Link]
- **Tong Wang (Tsinghua University) Research Group** — [Link]